

ISLAMI BANK BANGLADESH PLC

API HUB

Fee Collection API for All Channel

(CELLFIN - CBS - AGENT - MCASH)

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Version 3.2

Document Revision History

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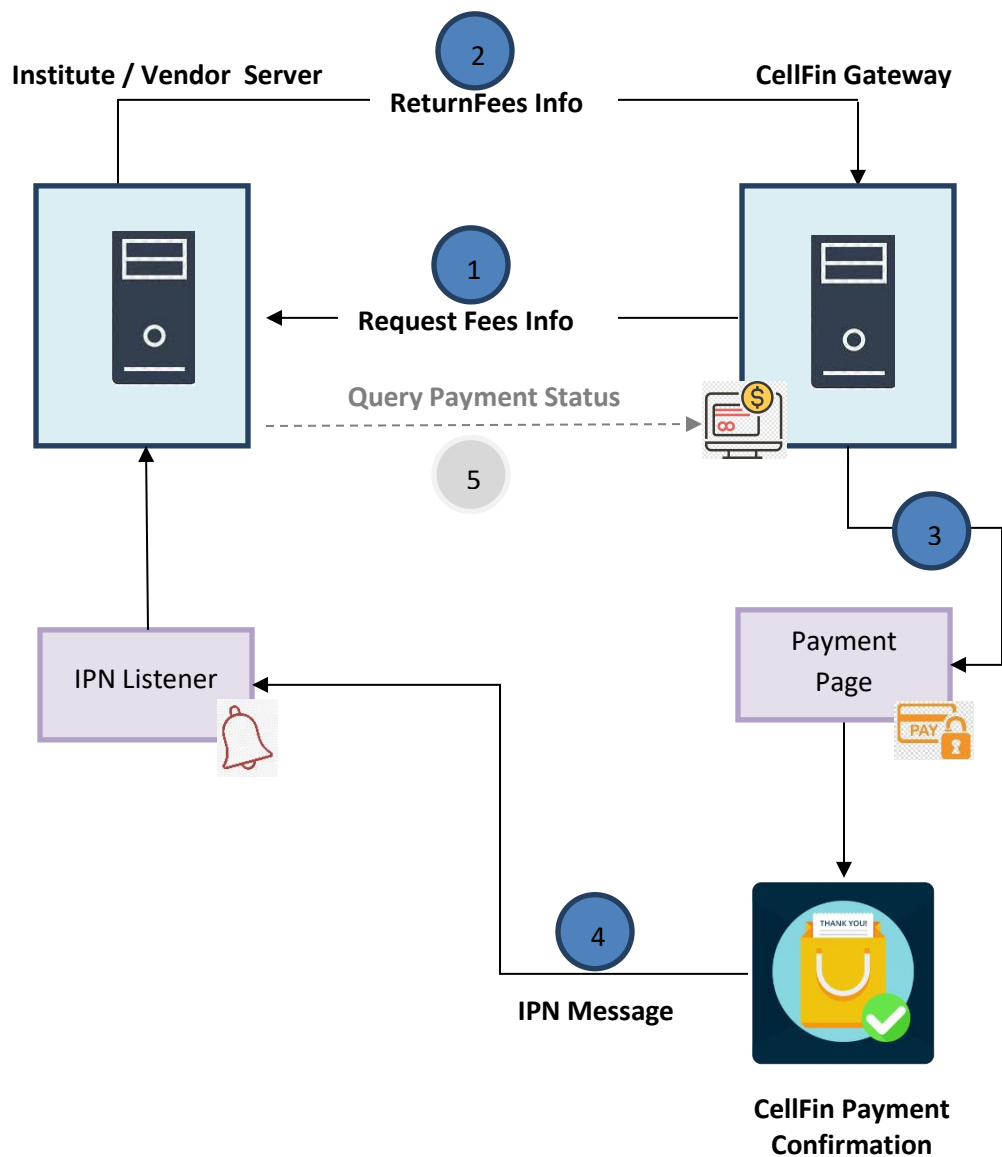
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1. Introduction

This document provides details specification for Fee Collection in CellFin/Cbs/Agent/mCash platforms. School/College/Universities can accept payment from Islami Bank customers through this APIs.

2. Transaction Process



There are three steps for payment integration.

Step 1 – IBBPLC sends query request to Vendor / Institute Server

Step 2 – Vendor / Institute Server returns fees info (If no fees are due, or the student ID / reference ID is invalid, the server will return an appropriate error code along with a clear error message. The returned message will be displayed directly to the user so they can understand the reason.)

Step 3 – IBBPLC system displays fees details to user. Upon user confirmation, the system debits the selected source account of the user. The corresponding amount is credited to the institute's account.

Step 4 – A payment confirmation message is sent to Vendor / Institute Server.

Step 5 – (Optional). Vendor / Institute Server can send a payment query request to IBBPLC server to check status of the payment.

2.1 Sample JSON fee info request

Step 1 & 2: Fee Info Request. Method HTTP POST. Content-Type is application/json

*** Request data must be JSON row data and ignore the other formats Like as :-
form-data ,x-www-form-urlencoded

Syntax of URL: <https://vendorserverip:port/v1/getFeeInfo>

```
{
  "userId": "IBBPLC_cf",
  "password":
    "a1f6dc7359f66d89236d7c3c2a2677bac7eeb7c78df3d84d74a90bc78cc63f72e5d690
    1a19ac9f7c958fb2a18328bada23e660073c866bb0573fa2d3f31a702a",
  "operation": "FEE_INFO",
  "iid": "iiuc",
  "referenceId": "ST1001",
  "random": "12323438765543"
  "billMonth": "012025"
}
```

userId – User Id given by vendor / institute to IBBPLC

iid– Institution ID given by vendor / institute to IBBPLC

referenceId – Student Id / Unique Id, user inputs for which payment is to be made

random – A random value that will be used as Salt value in password hash

password – SHA 512 hash of the password, random and referenceId (provided by vendor / institute) . e.g SHA512(password + random + referenceId)

operation – API operation e.g. FEE_INFO – for getting fee info

billMonth – (Optional) When an institute requires payments to be made on a monthly basis, this field may be used.

2.2 Sample JSON fee info response

Success	Failed
<pre>{ "referenceId": "ST1001", "dateTime": "22/08/2022 04:29 PM", "responseCode": "00", "responseMsg": "SUCCESS", "feeDetails": { "studentId": "ST1001", "instituteName": "School Name", "branchName": "Malibagh", "shift": "Morning", "className": "Six", "sectionName": "Beatles", "invoiceNo": "INV2208ST1001", "studentName": "Md. Ridwaan Ahmed", "fatherName": "Father Name", "month": "Aug", "academicYear": "2022", "fee": 500.00, "waiver": 0.00, "totalDue": 500.00 } }</pre>	<pre>{ "referenceId": "", "dateTime": "22/08/2022 04:29 PM ", "responseCode": "90", "responseMsg": "Data not found", "feeDetails": { "studentId": null, "instituteName": null, "branchName": null, "shift": null, "className": null, "sectionName": null, "invoiceNo": null, "studentName": null, "fatherName": null, "month": null, "academicYear": null, "fee": null, "waiver": null, "totalDue": null } }</pre>

referenceId – Should always match with the one sent in fee info request.

dateTime – Date and time of when the response was sent. Date format -

“yyyy-MM-dd'T'HH:mm:ss.SSS'Z”

responseCode – Denotes whether a referenceId was valid and has valid associated records like student name etc

responseMsg – Message that represent the responseCode meaning

feeDetails– contains fee details

studentId – Student Id, same as reference id

instituteName – Name of School/College/University

branchName – (Optional) If institute has different branches

shift – (Optional) shift value

className – Student’s class

sectionName – (Optional) Student’s section

invoiceNo– Invoice number of the payment. This is **mandatory** and will be used to reconcile any dispute.

studentName – Student Full Name

fatherName – Student’s Father Name

month – (Optional) Fee month.

academicYear – (Optional) Academic year

fee – Due fee amount. **Payment will be deducted based on this figure.**

Rest are for display only.

waiver – (Optional) if any

totalDue – Total due amount.

Step 3 & 4:

Fee information is displayed to user. If confirmed by user, payment is collected and deposited to Institute’s account. This is a parking account and school receives the fund at T+1 day in it’s settlement account. If payment is completed, an IPN notification is sent to Vendor / Institute server.

2.3 Sample JSON IPN notification

Following JSON notification is sent to IPN URL as soon as transaction is executed and whatever the outcome is. Method HTTP POST. Content-Type is application/json

*** Request data must be JSON row data and ignore the other formats Like as :- form-data ,x-www-form-urlencoded

Syntax of URL: <https://vendorserverip:port/v1/payment>

```
{
  "userId"           : "IBBPLCuser",
  "password"         :
    "a1f6dc7359f66d89236d7c3c2a2677bac7eeb7c78df3d84d74a90bc78cc63f72e5d690
    1a19ac9f7c958fb2a18328bada23e660073c866bb0573fa2d3f31a702a",
  "iid"              : "MKSH",
  "invoiceNo"        : "IVCF20251217123121261",
  "referenceId"      : "2511648",
  "trId"             : "8025121700***057",
  "status"           : "APPROVED",
  "date_time"        : "2025-12-17T12:31:48.895Z",
  "tr_amount"        : 226.0,
  "cr_amount"        : 220.0,
  "source_of_fund"   : "CELLFIN",
  "billMonth"        : "012025"
}
```

userId– User Id given by vendor/institute to IBBPLC

invoiceNo – The InvoiceNo was given by vendor/institute to IBBPLC by the response 2.2

iid– Institution ID given by vendor/institute to IBBPLC

password – SHA 512 hash of the password, trId (provided by IBBPLC) and invoiceNo

trId – Transaction ID provided by IBBPLC

tr_amount – Transaction amount is total debit amount. Institute fee and other fees (vat+tax etc).

cr_amount – Credit amount only Institute fee. Institute received this amount from student for institution.

date_time – Date format : “yyyy-MM-dd'T'HH:mm:ss.SSS'Z”

billMonth– When any institute wants monthly payment feature then they can use this field. Otherwise they can ignore.

2.3.1 Sample JSON IPN notification response

Success	Failed
<pre>{ "responseCode": "00", "responseMsg": "success" }</pre>	<pre>{ "responseCode": "90", "responseMsg": "Failed" }</pre>

2.4 Transaction Query API

Step 5: Vendor/Institute may send transaction query if a notification had not already been received via IPN, before updating its database and take any decision in terms of success/failure of the transaction.

Fee Info Request.

Method: HTTP POST.

Content-Type: application/json

*** Request data must be JSON row data and ignore the other formats Like as :-
form-data ,x-www-form-urlencoded

Syntax of URL: <https://staging.islamibankbd.com/api-hub/v2/getTrxInfo>

Basic Auth :

Username - test_user

Password - test_12345

Sample JSON transaction query request:

Header:

Content-Type: application/json

Authorization: Basic Y2VsbGZpbjpDM2xsZiFuMTI=

```
{
  "userId": "merchant.28201002",
  "password": "286f88c6eec7ae4a0c0d6fbc90d2751947ffd3df49ddaed25ce3386ccb4f7f1a5857f372ee9564d8f9aa216159909b8f7bed49742a3e4bc65753ca5d82c28da8",
  "operation": "STATUS",
  "random": "a28c6a6c-21e4-3c0d-a18b-43cee74ca787",
  "invoiceNo": "201025PAY10XAA"
}
```

userId– This User Id given by IBBPLC to vendor/institute

random– Random number is Transaction ID/trId. It's provided IBBPLC by the IPN notification to Vendor/institute

password – SHA 512 hash of the password, random (provided by IBBPLC) and invoiceNo

operation – API operation e.g. STATUS – to check transaction status

invoiceNo - The InvoiceNo was given by vendor/institute to IBBPLC

Sample JSON transaction query response:

Success	Failed
<pre>{ "invoiceNo": "201025PAY10XAA", "token": "a28c6a6c-21e4-3c0d-a18b-43cee74ca787-a52e53e0-1a61-42b7-86a1-342e137e1831", "trId": "3220112000000001", "status": "APPROVED", "date_time": "20/11/2020 04:29 PM", "tr_amount": 110.0, "cr_amount": 100.0, "source_of_fund": "CELLFIN" }</pre>	<pre>{ "status": "NOT_FOUND", "tr_amount": 0.0, "cr_amount": 0.0 }</pre>

2.5 Request Parameters

Name	Type	Length	Description
Fee Info Request			
userId	String (required)	32	ID provided by institution/Vendor
password	String (required)	200	SHA 512 hash of the password (provided by vendor) +random+referenceId
operation	String (required)	50	FEE_INFO (API operation for getting fee info)
iid	String (required)	16	Institute Id (provided by vendor)
referenceId	String (required)	32	Unique id student in vendor system
feeDetails			
studentId	String (required)	32	Unique id student in vendor system
instituteName	String (required)	50	
branchName	String	50	
shift	String	16	
className	String (required)	16	
sectionName	String	30	
invoiceNo	String (required)	32	
studentName	String (required)	50	
fatherName	String	50	
month	String	12	
academicYear	String	12	
fee	Decimal (required)	10,2 (e.g. 200.00)	
waiver	Decimal (required)	10,2 (e.g. 200.00)	
totalDue	Decimal (required)	10,2 (e.g. 200.00)	
dateTime	String		yyyy-MM-dd'T'HH:mm:ss.SSS'Z'
IPN notification			
invoiceNo	String	50	Unique id from merchant system to identify this transaction
password	String	200	SHA 512 hash of the password (provided by vendor) +trId+invoiceNo
trId	String		Transaction ID
status	String	50	APPROVED(Status of the transaction)
date_time	String		yyyy-MM-dd'T'HH:mm:ss.SSS'Z'

tr_amount	Decimal	10,2 (e.g. 200.00)	Transaction amount
cr_amount	Decimal	10,2 (e.g. 200.00)	Credit amount
source of fund	String		Source of fund e.g. CELLFIN
Transaction Query			
userId	String	32	ID provided by institution/Vendor
password	String	100	SHA 512 hash of the password (providedby vendor) +random+ invoiceNo
operation	String	Fixed String	STATUS (Status of the transaction)
random	String	200	
invoiceNo	String	50	Unique id from merchant system to identify this transaction

Response Status Legends

QUEUED	Gateway received the transaction request
OTP_SENT	Transaction created, and validation OTP sent to customer
APPROVED	Transaction successful
FAILED	Transaction failed
CANCELLED	Customer canceled transaction
UN_ATTEMPTED	Customer did not attempt the transaction
EXCEED_FUND_AVAILABLE	Not enough credit in customer account for this transaction
DUPLICATE_TOKEN_REQUEST	A token request for transaction with same sessionId already exists
EXCEED_DAILY_LIMIT	Daily transaction limit reached
EXCEED_MONTHLY_LIMIT	Monthly transaction limit reached
INVALID_TRANSACTION	Requested transaction invalid. e.g. redirect URL contains invalid token or already used token
EXPIRED	Transaction expired (token creation time and transaction time cannot be more than 10 minutes)
REFUNDED	Refund executed
UNABLE_TO_PROCESS	Internal error. System failed to process the request
UNABLE_TO_REFUND	Refund failed due to either fund unavailability or system error
AUTHENTICATION_FAILED	Hashed password does not match.
NOT_FOUND	Request transaction (in case of App in App session id) with given details does not exist.